

TrES-2b

R-0164

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_180827 · created 2026-07-16T16:23:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458357.79887 +/- 0.00061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.089 +/- 0.039
Transit depth (Rp/Rs)^2	0.79 +/- 0.69 [%]
Orbital Inclination (inc)	83.48 +/- 2.55
Airmass coefficient 1 (a1)	1.283 +/- 0.014
Airmass coefficient 2 (a2)	-0.1753 +/- 0.0074
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	None
Optimal Aperture	4.53
Optimal Annulus	9.41
Transit Duration (day)	0.051 +/- 0.04

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.089 ± 0.039 vs 0.1254 (deviation 0.85σ)
Duration fit vs published	0.051 d vs 0.0746 d (deviation 0.53σ)
Mid-time O–C	1.3 min
Depth SNR	2.1
Residual scatter	1.07 %

Light curve

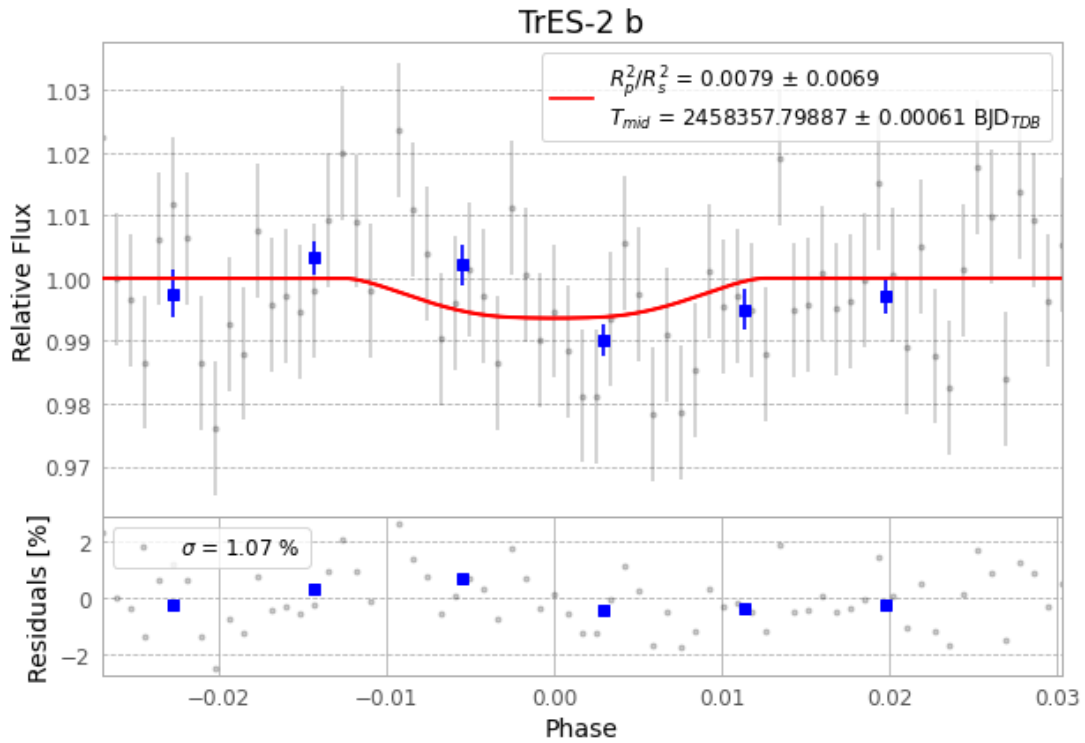


Figure 1 — FinalLightCurve_TrES-2 b_2018-08-26.png (SHA-256 235966bf035dbfd0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [335, 309], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 547c7d2abf7dd256...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1634555

Data provenance

69 FITS frames (45 MB), TRES-2180827053013.FITS → TRES-2180827085412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-180827.log (874b9d3427ea...), FinalLightCurve_TrES-2 b_2018-08-26.png (235966bf035d...), FinalLightCurve_TrES-2 b_2018-08-26.csv (600bbcb7f8a4...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 16:23Z.